

EquiAudit Fairness Report

Stage 4: Base Fairness ML Model

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood	4689	0.8454	0.7641	0.2329	0.1802	0.4451	0.0664
Elderly	890	0.8539	0.7842	0.2438	0.1876	0.4147	0.0594
Midlife	4182	0.7896	0.7698	0.3757	0.3312	0.3393	0.1329
Youth	2450	0.9857	0.6662	0.0171	0.0045	0.7857	0.0008

Detailed Group Metrics: Workclass

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Self-emp-not-inc	971	0.8198	0.7488	0.2750	0.1936	0.4757	0.0682
Private	8482	0.8606	0.7831	0.2213	0.1807	0.4065	0.0634
Self-emp-inc	411	0.7956	0.7922	0.5766	0.5523	0.1983	0.2126
Local-gov	744	0.8105	0.7478	0.2621	0.2392	0.4051	0.1129
Federal-gov	350	0.8257	0.8148	0.3886	0.3686	0.2500	0.1262
State-gov	535	0.8673	0.8142	0.2636	0.2019	0.3688	0.0482
Missing	713	0.9215	0.7330	0.0968	0.0631	0.5797	0.0248
Without-pay	3	1.0000	1.0000	0.0000	0.0000	0	0.0000
Never-worked	2	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
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Bachelors	2071	0.8093	0.8024	0.4153	0.3984	0.2500	0.1486
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8644	0.6929	0.1576	0.0952	0.6282	0.0435
Assoc-voc	497	0.8089	0.7353	0.2656	0.2072	0.4697	0.0904
Prof-school	224	0.8705	0.8358	0.7009	0.7589	0.0510	0.3134
10th	355	0.9352	0.6880	0.0789	0.0310	0.7143	0.0092
Some-college	2683	0.8565	0.7357	0.1826	0.1413	0.5061	0.0625
7th-8th	249	0.9157	0.5578	0.0884	0.0120	0.9091	0.0044
Masters	637	0.8038	0.8025	0.5589	0.5228	0.2079	0.1815
11th	442	0.9593	0.6895	0.0475	0.0204	0.7143	0.0071
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8153	0.7277	0.2427	0.1900	0.4891	0.0871
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7355	0.6851	0.6581	0.7419	0.1373	0.5094
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9705	0.7580	0.0440	0.0189	0.6201	0.0023
Married-civ-spouse	5567	0.7358	0.7296	0.4489	0.3999	0.3489	0.1952
Divorced	1682	0.9287	0.7244	0.0957	0.0434	0.6460	0.0105
Separated	373	0.9464	0.7585	0.0777	0.0402	0.5862	0.0087
Married-spouse-absent	152	0.9276	0.7603	0.1184	0.0461	0.6111	0.0000
Widowed	359	0.9359	0.7025	0.0864	0.0279	0.7097	0.0030
Married-AF-spouse	13	0.6154	0.3810	0.3846	0.0000	1.0000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
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Other-service	1248	0.9599	0.6221	0.0441	0.0104	0.8364	0.0034
Farming-fishing	370	0.8892	0.7095	0.1243	0.0892	0.5870	0.0432
Sales	1368	0.8311	0.7677	0.2617	0.2156	0.4106	0.0832
Exec-managerial	1472	0.7962	0.7960	0.5061	0.4599	0.2470	0.1596
Craft-repair	1564	0.8024	0.6873	0.2212	0.1720	0.5578	0.0952
Prof-specialty	1568	0.8138	0.8081	0.4298	0.3980	0.2537	0.1353
Machine-oper-inspct	727	0.8624	0.6410	0.1486	0.0660	0.7407	0.0323
Tech-support	394	0.8223	0.7700	0.3020	0.2208	0.4286	0.0691
Adm-clerical	1413	0.9016	0.7705	0.1331	0.1111	0.4521	0.0441
Missing	715	0.9217	0.7331	0.0965	0.0629	0.5797	0.0248
Handlers-cleaners	514	0.9475	0.7314	0.0700	0.0331	0.6389	0.0084
Transport-moving	576	0.8281	0.6893	0.1875	0.1441	0.5741	0.0791
Protective-serv	229	0.7642	0.7014	0.2926	0.2489	0.4776	0.1358
Priv-house-serv	52	0.9615	0.8231	0.0577	0.0577	0.3333	0.0204
Armed-Forces	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9916	0.7626	0.0126	0.0053	0.6250	0.0005
Husband	4916	0.7325	0.7266	0.4530	0.3999	0.3538	0.1960
Not-in-family	3148	0.9320	0.7450	0.0985	0.0451	0.6161	0.0081
Unmarried	1311	0.9504	0.7044	0.0648	0.0229	0.7059	0.0041
Wife	580	0.7431	0.7402	0.4517	0.4431	0.2939	0.2264
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
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White	10442	0.8468	0.7860	0.2547	0.2124	0.3838	0.0744
Black	1186	0.9224	0.7868	0.1214	0.0809	0.4861	0.0211
Amer-Indian-Eskimo	126	0.9127	0.7369	0.0873	0.0952	0.4545	0.0522
Asian-Pac-Islander	369	0.8401	0.7755	0.2602	0.2033	0.4167	0.0696
Other	88	0.9091	0.7525	0.1250	0.0795	0.5455	0.0260

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9300	0.7995	0.1117	0.0816	0.4481	0.0225
Male	8154	0.8178	0.7734	0.3028	0.2547	0.3803	0.0962

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8526	0.7873	0.2430	0.2029	0.3856	0.0709
Ireland	12	0.8333	0.8125	0.4167	0.2500	0.4000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8708	0.8283	0.2727	0.2297	0.3158	0.0592
Puerto-Rico	49	0.9592	0.8639	0.0816	0.0816	0.2500	0.0222
Philippines	73	0.8493	0.7633	0.2192	0.1781	0.4375	0.0702
China	32	0.9062	0.8798	0.3125	0.2188	0.3000	0.0000
Germany	54	0.8889	0.8393	0.2593	0.1852	0.3571	0.0250
Mexico	211	0.9668	0.6026	0.0284	0.0142	0.8333	0.0098
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7500	0.7402	0.4722	0.3333	0.4118	0.1053
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8000	0.7964	0.4667	0.4000	0.2857	0.1250
Vietnam	23	0.9130	0.7262	0.0435	0.1304	0.0000	0.0909
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.4286	0.2500	0.0000	1.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000

Greece	14	0.5714	0.5625	0.5714	0.2857	0.6250	0.1667
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7708	0.7166	0.3542	0.2083	0.5294	0.0645
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7895	0.7286	0.3421	0.1842	0.5385	0.0400
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI- tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9500	0.8857	0.1000	0.1500	0.0000	0.0556
Italy	28	0.7500	0.6813	0.3571	0.1786	0.6000	0.0556
Trinidad&To bago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	1.0000	1.0000	0.0526	0.0526	0.0000	0.0000
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.5714	0.5333	0.5714	0.1429	0.7500	0.0000
Japan	23	0.8696	0.8392	0.3043	0.2609	0.2857	0.0625
Scotland	5	0.8000	0.4444	0.2000	0.0000	1.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Stage 4.5: Per-Attribute Fairness ML Model

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood	4689	0.8454	0.7641	0.2329	0.1802	0.4451	0.0664
Elderly	890	0.8539	0.7842	0.2438	0.1876	0.4147	0.0594
Midlife	4182	0.7896	0.7698	0.3757	0.3312	0.3393	0.1329
Youth	2450	0.9857	0.6662	0.0171	0.0045	0.7857	0.0008

Detailed Group Metrics: Workclass

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Self-emp-not-inc	971	0.8198	0.7488	0.2750	0.1936	0.4757	0.0682
Private	8482	0.8606	0.7831	0.2213	0.1807	0.4065	0.0634
Self-emp-inc	411	0.7956	0.7922	0.5766	0.5523	0.1983	0.2126
Local-gov	744	0.8105	0.7478	0.2621	0.2392	0.4051	0.1129
Federal-gov	350	0.8257	0.8148	0.3886	0.3686	0.2500	0.1262
State-gov	535	0.8673	0.8142	0.2636	0.2019	0.3688	0.0482
Missing	713	0.9215	0.7330	0.0968	0.0631	0.5797	0.0248
Without-pay	3	1.0000	1.0000	0.0000	0.0000	0	0.0000
Never-worked	2	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8093	0.8024	0.4153	0.3984	0.2500	0.1486
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8644	0.6929	0.1576	0.0952	0.6282	0.0435
Assoc-voc	497	0.8089	0.7353	0.2656	0.2072	0.4697	0.0904
Prof-school	224	0.8705	0.8358	0.7009	0.7589	0.0510	0.3134
10th	355	0.9352	0.6880	0.0789	0.0310	0.7143	0.0092
Some-college	2683	0.8565	0.7357	0.1826	0.1413	0.5061	0.0625
7th-8th	249	0.9157	0.5578	0.0884	0.0120	0.9091	0.0044
Masters	637	0.8038	0.8025	0.5589	0.5228	0.2079	0.1815

11th	442	0.9593	0.6895	0.0475	0.0204	0.7143	0.0071
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8153	0.7277	0.2427	0.1900	0.4891	0.0871
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7355	0.6851	0.6581	0.7419	0.1373	0.5094
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9705	0.7580	0.0440	0.0189	0.6201	0.0023
Married-civ-spouse	5567	0.7358	0.7296	0.4489	0.3999	0.3489	0.1952
Divorced	1682	0.9287	0.7244	0.0957	0.0434	0.6460	0.0105
Separated	373	0.9464	0.7585	0.0777	0.0402	0.5862	0.0087
Married-spouse-absent	152	0.9276	0.7603	0.1184	0.0461	0.6111	0.0000
Widowed	359	0.9359	0.7025	0.0864	0.0279	0.7097	0.0030
Married-AF-spouse	13	0.6154	0.3810	0.3846	0.0000	1.0000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9599	0.6221	0.0441	0.0104	0.8364	0.0034
Farming-fishing	370	0.8892	0.7095	0.1243	0.0892	0.5870	0.0432
Sales	1368	0.8311	0.7677	0.2617	0.2156	0.4106	0.0832
Exec-managerial	1472	0.7962	0.7960	0.5061	0.4599	0.2470	0.1596
Craft-repair	1564	0.8024	0.6873	0.2212	0.1720	0.5578	0.0952
Prof-specialty	1568	0.8138	0.8081	0.4298	0.3980	0.2537	0.1353
Machine-oper-inspct	727	0.8624	0.6410	0.1486	0.0660	0.7407	0.0323
Tech-support	394	0.8223	0.7700	0.3020	0.2208	0.4286	0.0691

Adm-clerical	1413	0.9016	0.7705	0.1331	0.1111	0.4521	0.0441
Missing	715	0.9217	0.7331	0.0965	0.0629	0.5797	0.0248
Handlers-cle aners	514	0.9475	0.7314	0.0700	0.0331	0.6389	0.0084
Transport-mo ving	576	0.8281	0.6893	0.1875	0.1441	0.5741	0.0791
Protective-ser v	229	0.7642	0.7014	0.2926	0.2489	0.4776	0.1358
Priv-house-se rv	52	0.9615	0.8231	0.0577	0.0577	0.3333	0.0204
Armed-Force s	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9916	0.7626	0.0126	0.0053	0.6250	0.0005
Husband	4916	0.7325	0.7266	0.4530	0.3999	0.3538	0.1960
Not-in-family	3148	0.9320	0.7450	0.0985	0.0451	0.6161	0.0081
Unmarried	1311	0.9504	0.7044	0.0648	0.0229	0.7059	0.0041
Wife	580	0.7431	0.7402	0.4517	0.4431	0.2939	0.2264
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8468	0.7860	0.2547	0.2124	0.3838	0.0744
Black	1186	0.9224	0.7868	0.1214	0.0809	0.4861	0.0211
Amer-Indian-Eskimo	126	0.9127	0.7369	0.0873	0.0952	0.4545	0.0522
Asian-Pac-Isl ander	369	0.8401	0.7755	0.2602	0.2033	0.4167	0.0696
Other	88	0.9091	0.7525	0.1250	0.0795	0.5455	0.0260

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
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Female	4057	0.9300	0.7995	0.1117	0.0816	0.4481	0.0225
Male	8154	0.8178	0.7734	0.3028	0.2547	0.3803	0.0962

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8526	0.7873	0.2430	0.2029	0.3856	0.0709
Ireland	12	0.8333	0.8125	0.4167	0.2500	0.4000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8708	0.8283	0.2727	0.2297	0.3158	0.0592
Puerto-Rico	49	0.9592	0.8639	0.0816	0.0816	0.2500	0.0222
Philippines	73	0.8493	0.7633	0.2192	0.1781	0.4375	0.0702
China	32	0.9062	0.8798	0.3125	0.2188	0.3000	0.0000
Germany	54	0.8889	0.8393	0.2593	0.1852	0.3571	0.0250
Mexico	211	0.9668	0.6026	0.0284	0.0142	0.8333	0.0098
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7500	0.7402	0.4722	0.3333	0.4118	0.1053
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8000	0.7964	0.4667	0.4000	0.2857	0.1250
Vietnam	23	0.9130	0.7262	0.0435	0.1304	0.0000	0.0909
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.4286	0.2500	0.0000	1.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.5714	0.5625	0.5714	0.2857	0.6250	0.1667
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7708	0.7166	0.3542	0.2083	0.5294	0.0645
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7895	0.7286	0.3421	0.1842	0.5385	0.0400
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000

Outlying-US(Guam-USVI-etc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-Republic	20	0.9500	0.8857	0.1000	0.1500	0.0000	0.0556
Italy	28	0.7500	0.6813	0.3571	0.1786	0.6000	0.0556
Trinidad&Tobago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	1.0000	1.0000	0.0526	0.0526	0.0000	0.0000
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.5714	0.5333	0.5714	0.1429	0.7500	0.0000
Japan	23	0.8696	0.8392	0.3043	0.2609	0.2857	0.0625
Scotland	5	0.8000	0.4444	0.2000	0.0000	1.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9264	0.8035	0.1202	0.0889	0.4362	0.0241
Black + Male	600	0.8833	0.7733	0.1783	0.1250	0.4766	0.0385
White + Male	7181	0.8126	0.7742	0.3158	0.2721	0.3660	0.1050
Black + Female	586	0.9556	0.7851	0.0631	0.0461	0.4865	0.0146
Amer-Indian-Eskimo + Male	73	0.9041	0.7045	0.0685	0.1096	0.4000	0.0735
Asian-Pac-Islander + Female	131	0.9237	0.7519	0.1221	0.0458	0.6250	0.0000
Amer-Indian-Eskimo + Female	53	0.9057	0.6964	0.1132	0.0566	0.6667	0.0213
Asian-Pac-Islander + Male	238	0.7899	0.7530	0.3361	0.2773	0.4000	0.1139

Other + Female	26	0.9231	0.7292	0.0769	0.0769	0.5000	0.0417
Other + Male	62	0.9032	0.7584	0.1452	0.0806	0.5556	0.0189

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood + Female	1505	0.9090	0.7864	0.1389	0.1037	0.4545	0.0324
Early Adulthood + Male	3184	0.8144	0.7505	0.2773	0.2167	0.4439	0.0865
Elderly + Male	609	0.8013	0.7553	0.3169	0.2496	0.4197	0.0962
Midlife + Female	1188	0.8956	0.7970	0.1759	0.1271	0.4354	0.0337
Youth + Female	1083	0.9926	0.7481	0.0102	0.0046	0.6364	0.0009
Youth + Male	1367	0.9810	0.6531	0.0227	0.0051	0.8065	0.0007
Elderly + Female	281	0.9502	0.8116	0.0854	0.0569	0.4583	0.0117
Midlife + Male	2994	0.7525	0.7486	0.4549	0.4205	0.3098	0.1955

Stage 4.5: Intersectional Fairness ML Model

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9264	0.8035	0.1202	0.0889	0.4362	0.0241
Black + Male	600	0.8833	0.7733	0.1783	0.1250	0.4766	0.0385
White + Male	7181	0.8126	0.7742	0.3158	0.2721	0.3660	0.1050
Black + Female	586	0.9556	0.7851	0.0631	0.0461	0.4865	0.0146
Amer-Indian-Eskimo + Male	73	0.9041	0.7045	0.0685	0.1096	0.4000	0.0735
Asian-Pac-Islander + Female	131	0.9237	0.7519	0.1221	0.0458	0.6250	0.0000

Amer-Indian-Eskimo + Female	53	0.9057	0.6964	0.1132	0.0566	0.6667	0.0213
Asian-Pac-Islander + Male	238	0.7899	0.7530	0.3361	0.2773	0.4000	0.1139
Other + Female	26	0.9231	0.7292	0.0769	0.0769	0.5000	0.0417
Other + Male	62	0.9032	0.7584	0.1452	0.0806	0.5556	0.0189

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood + Female	1505	0.9090	0.7864	0.1389	0.1037	0.4545	0.0324
Early Adulthood + Male	3184	0.8144	0.7505	0.2773	0.2167	0.4439	0.0865
Elderly + Male	609	0.8013	0.7553	0.3169	0.2496	0.4197	0.0962
Midlife + Female	1188	0.8956	0.7970	0.1759	0.1271	0.4354	0.0337
Youth + Female	1083	0.9926	0.7481	0.0102	0.0046	0.6364	0.0009
Youth + Male	1367	0.9810	0.6531	0.0227	0.0051	0.8065	0.0007
Elderly + Female	281	0.9502	0.8116	0.0854	0.0569	0.4583	0.0117
Midlife + Male	2994	0.7525	0.7486	0.4549	0.4205	0.3098	0.1955

Stage 6: Post-Mitigation ML Model (Reweighting)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood	4689	0.8334	0.7499	0.2329	0.1892	0.4515	0.0801
Elderly	890	0.8483	0.7788	0.2438	0.1955	0.4101	0.0684
Midlife	4182	0.7834	0.7624	0.3757	0.3274	0.3526	0.1348
Youth	2450	0.9841	0.6960	0.0171	0.0094	0.6905	0.0042

Detailed Group Metrics: Workclass

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Self-emp-not-inc	971	0.8126	0.7460	0.2750	0.2132	0.4532	0.0866
Private	8482	0.8540	0.7752	0.2213	0.1864	0.4086	0.0713
Self-emp-inc	411	0.7737	0.7703	0.5766	0.5450	0.2236	0.2299
Local-gov	744	0.8105	0.7450	0.2621	0.2312	0.4205	0.1075
Federal-gov	350	0.8057	0.7853	0.3886	0.3029	0.3603	0.0888
State-gov	535	0.8598	0.8037	0.2636	0.2019	0.3830	0.0533
?	713	0.9102	0.7174	0.0968	0.0771	0.5652	0.0388
Without-pay	3	1.0000	1.0000	0.0000	0.0000	0	0.0000
Never-worked	2	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.7861	0.7753	0.4153	0.3655	0.3174	0.1404
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8586	0.7073	0.1576	0.1234	0.5570	0.0636
Assoc-voc	497	0.8189	0.7528	0.2656	0.2173	0.4318	0.0904
Prof-school	224	0.8571	0.8233	0.7009	0.7366	0.0764	0.2985
10th	355	0.9211	0.6608	0.0789	0.0451	0.7143	0.0245
Some-college	2683	0.8565	0.7428	0.1826	0.1524	0.4755	0.0693
7th-8th	249	0.9197	0.5942	0.0884	0.0161	0.8636	0.0044
Masters	637	0.7865	0.7861	0.5589	0.4835	0.2584	0.1566
11th	442	0.9570	0.7010	0.0475	0.0271	0.6667	0.0119
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8021	0.7043	0.2427	0.1821	0.5326	0.0906
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7290	0.6961	0.6581	0.6710	0.1961	0.4151
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9668	0.7423	0.0440	0.0226	0.6201	0.0062
Married-civilian spouse	5567	0.7230	0.7168	0.4489	0.4029	0.3597	0.2096
Divorced	1682	0.9257	0.7249	0.0957	0.0499	0.6273	0.0158
Separated	373	0.9437	0.7517	0.0777	0.0429	0.5862	0.0116
Married-spouse-absent	152	0.9276	0.7603	0.1184	0.0461	0.6111	0.0000
Widowed	359	0.9359	0.7025	0.0864	0.0279	0.7097	0.0030
Married-AF-spouse	13	0.6923	0.5667	0.3846	0.0769	0.8000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9519	0.6542	0.0441	0.0280	0.7273	0.0168
Farming-fishing	370	0.8486	0.6649	0.1243	0.1351	0.5652	0.0926
Sales	1368	0.8282	0.7632	0.2617	0.2142	0.4190	0.0842
Exec-managerial	1472	0.7758	0.7749	0.5061	0.4314	0.2953	0.1513
Craft-repair	1564	0.8024	0.6996	0.2212	0.1937	0.5087	0.1092
Prof-specialty	1568	0.8048	0.7970	0.4298	0.3737	0.2923	0.1219
Machine-oper-inspct	727	0.8556	0.6718	0.1486	0.1032	0.6389	0.0582
Tech-support	394	0.8249	0.7633	0.3020	0.1878	0.4790	0.0436
Adm-clerical	1413	0.8946	0.7490	0.1331	0.1054	0.5000	0.0449
?	715	0.9105	0.7174	0.0965	0.0769	0.5652	0.0387
Handlers-cleaners	514	0.9358	0.7290	0.0700	0.0564	0.5556	0.0272
Transport-moving	576	0.8351	0.7224	0.1875	0.1753	0.4722	0.0940
Protective-service	229	0.7860	0.7404	0.2926	0.2882	0.3731	0.1481
Priv-house-serv	52	0.9808	0.8949	0.0577	0.0385	0.3333	0.0000

Armed-Forces	1	1.0000	1.0000	0.0000	0.0000	0	0.0000
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Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9905	0.7476	0.0126	0.0063	0.6250	0.0016
Husband	4916	0.7217	0.7162	0.4530	0.4074	0.3574	0.2127
Not-in-family	3148	0.9263	0.7353	0.0985	0.0521	0.6097	0.0152
Unmarried	1311	0.9497	0.7072	0.0648	0.0252	0.6941	0.0057
Wife	580	0.7138	0.7082	0.4517	0.4103	0.3626	0.2233
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8391	0.7765	0.2547	0.2161	0.3917	0.0820
Black	1186	0.9182	0.7760	0.1214	0.0818	0.5000	0.0240
Amer-Indian-Eskimo	126	0.8968	0.7114	0.0873	0.1111	0.4545	0.0696
Asian-Pac-Islander	369	0.8266	0.7594	0.2602	0.2114	0.4271	0.0842
Other	88	0.9205	0.7935	0.1250	0.0909	0.4545	0.0260

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9236	0.7786	0.1117	0.0791	0.4879	0.0247
Male	8154	0.8098	0.7652	0.3028	0.2615	0.3823	0.1068

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8448	0.7776	0.2430	0.2073	0.3927	0.0789
Ireland	12	0.7500	0.6975	0.4167	0.1667	0.6000	0.0000

Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
?	209	0.8660	0.8208	0.2727	0.2249	0.3333	0.0592
Puerto-Rico	49	0.9796	0.9231	0.0816	0.0612	0.2500	0.0000
Philippines	73	0.8219	0.7203	0.2192	0.1781	0.5000	0.0877
China	32	0.8438	0.7997	0.3125	0.2188	0.4000	0.0455
Germany	54	0.8704	0.8272	0.2593	0.2407	0.2857	0.0750
Mexico	211	0.9716	0.6178	0.0284	0.0095	0.8333	0.0049
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.6667	0.6625	0.4722	0.4167	0.4118	0.2632
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8667	0.8611	0.4667	0.3333	0.2857	0.0000
Vietnam	23	0.9565	0.8217	0.0435	0.0870	0.0000	0.0455
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.4286	0.2500	0.0000	1.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.5000	0.4974	0.5714	0.3571	0.6250	0.3333
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7292	0.6485	0.3542	0.1667	0.6471	0.0645
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7895	0.7286	0.3421	0.1842	0.5385	0.0400
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI-e tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9000	0.7222	0.1000	0.1000	0.5000	0.0556
Italy	28	0.7857	0.7375	0.3571	0.2143	0.5000	0.0556
Trinidad&To bago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.7857	0.6348	0.2143	0.1429	0.6667	0.0909

Columbia	19	1.0000	1.0000	0.0526	0.0526	0.0000	0.0000
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.9130	0.8873	0.3043	0.2174	0.2857	0.0000
Scotland	5	0.8000	0.4444	0.2000	0.0000	1.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9175	0.7777	0.1202	0.0868	0.4821	0.0279
Black + Male	600	0.8733	0.7493	0.1783	0.1183	0.5234	0.0406
White + Male	7181	0.8035	0.7639	0.3158	0.2748	0.3761	0.1136
Black + Female	586	0.9642	0.8239	0.0631	0.0444	0.4324	0.0091
Amer-Indian-Eskimo + Male	73	0.8904	0.7192	0.0685	0.1507	0.2000	0.1029
Asian-Pac-Islander + Female	131	0.9008	0.6902	0.1221	0.0534	0.6875	0.0174
Amer-Indian-Eskimo + Female	53	0.9057	0.6964	0.1132	0.0566	0.6667	0.0213
Asian-Pac-Islander + Male	238	0.7857	0.7527	0.3361	0.2983	0.3750	0.1329
Other + Female	26	0.9231	0.7292	0.0769	0.0769	0.5000	0.0417
Other + Male	62	0.9194	0.8104	0.1452	0.0968	0.4444	0.0189

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood + Female	1505	0.8950	0.7543	0.1389	0.1043	0.5024	0.0409

Early Adulthood + Male	3184	0.8043	0.7414	0.2773	0.2293	0.4394	0.1021
Elderly + Male	609	0.7997	0.7563	0.3169	0.2611	0.4041	0.1058
Midlife + Female	1188	0.8897	0.7821	0.1759	0.1212	0.4689	0.0337
Youth + Female	1083	0.9926	0.7481	0.0102	0.0046	0.6364	0.0009
Youth + Male	1367	0.9773	0.6779	0.0227	0.0132	0.7097	0.0067
Elderly + Female	281	0.9537	0.8209	0.0854	0.0534	0.4583	0.0078
Midlife + Male	2994	0.7411	0.7363	0.4549	0.4092	0.3348	0.1955

Comparative Group Metrics: Before vs After (Reweighting)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early Adulthood	0.7641	0.7499	0.4451	0.4515	0.0664	0.0801	0.1802	0.1892
Elderly	0.7842	0.7788	0.4147	0.4101	0.0594	0.0684	0.1876	0.1955
Midlife	0.7698	0.7624	0.3393	0.3526	0.1329	0.1348	0.3312	0.3274
Youth	0.6662	0.6960	0.7857	0.6905	0.0008	0.0042	0.0045	0.0094

Workclass

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Self-emp-not-inc	0.7488	0.7460	0.4757	0.4532	0.0682	0.0866	0.1936	0.2132
Private	0.7831	0.7752	0.4065	0.4086	0.0634	0.0713	0.1807	0.1864
Self-emp-inc	0.7922	0.7703	0.1983	0.2236	0.2126	0.2299	0.5523	0.5450
Local-gov	0.7478	0.7450	0.4051	0.4205	0.1129	0.1075	0.2392	0.2312
Federal-gov	0.8148	0.7853	0.2500	0.3603	0.1262	0.0888	0.3686	0.3029
State-gov	0.8142	0.8037	0.3688	0.3830	0.0482	0.0533	0.2019	0.2019
Missing	0.7330	N/A	0.5797	N/A	0.0248	N/A	0.0631	N/A
Without-pay	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Never-worked	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.8024	0.7753	0.2500	0.3174	0.1486	0.1404	0.3984	0.3655
9th	0.4918	0.4918	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
HS-grad	0.6929	0.7073	0.6282	0.5570	0.0435	0.0636	0.0952	0.1234
Assoc-voc	0.7353	0.7528	0.4697	0.4318	0.0904	0.0904	0.2072	0.2173
Prof-school	0.8358	0.8233	0.0510	0.0764	0.3134	0.2985	0.7589	0.7366
10th	0.6880	0.6608	0.7143	0.7143	0.0092	0.0245	0.0310	0.0451
Some-college	0.7357	0.7428	0.5061	0.4755	0.0625	0.0693	0.1413	0.1524
7th-8th	0.5578	0.5942	0.9091	0.8636	0.0044	0.0044	0.0120	0.0161
Masters	0.8025	0.7861	0.2079	0.2584	0.1815	0.1566	0.5228	0.4835
11th	0.6895	0.7010	0.7143	0.6667	0.0071	0.0119	0.0204	0.0271
5th-6th	0.4762	0.4762	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Assoc-acadm	0.7277	0.7043	0.4891	0.5326	0.0871	0.0906	0.1900	0.1821
Preschool	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0455	0.0455
12th	0.7640	0.7640	0.6154	0.6154	0.0000	0.0000	0.0325	0.0325
Doctorate	0.6851	0.6961	0.1373	0.1961	0.5094	0.4151	0.7419	0.6710
1st-4th	0.4945	0.4945	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7580	0.7423	0.6201	0.6201	0.0023	0.0062	0.0189	0.0226
Married-civ-spouse	0.7296	0.7168	0.3489	0.3597	0.1952	0.2096	0.3999	0.4029
Divorced	0.7244	0.7249	0.6460	0.6273	0.0105	0.0158	0.0434	0.0499
Separated	0.7585	0.7517	0.5862	0.5862	0.0087	0.0116	0.0402	0.0429
Married-spouse-absent	0.7603	0.7603	0.6111	0.6111	0.0000	0.0000	0.0461	0.0461
Widowed	0.7025	0.7025	0.7097	0.7097	0.0030	0.0030	0.0279	0.0279
Married-AF-spouse	0.3810	0.5667	1.0000	0.8000	0.0000	0.0000	0.0000	0.0769

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6221	0.6542	0.8364	0.7273	0.0034	0.0168	0.0104	0.0280
Farming-fishing	0.7095	0.6649	0.5870	0.5652	0.0432	0.0926	0.0892	0.1351
Sales	0.7677	0.7632	0.4106	0.4190	0.0832	0.0842	0.2156	0.2142
Exec-managerial	0.7960	0.7749	0.2470	0.2953	0.1596	0.1513	0.4599	0.4314
Craft-repair	0.6873	0.6996	0.5578	0.5087	0.0952	0.1092	0.1720	0.1937
Prof-specialty	0.8081	0.7970	0.2537	0.2923	0.1353	0.1219	0.3980	0.3737
Machine-op-inspct	0.6410	0.6718	0.7407	0.6389	0.0323	0.0582	0.0660	0.1032
Tech-support	0.7700	0.7633	0.4286	0.4790	0.0691	0.0436	0.2208	0.1878
Adm-clerical	0.7705	0.7490	0.4521	0.5000	0.0441	0.0449	0.1111	0.1054
Missing	0.7331	N/A	0.5797	N/A	0.0248	N/A	0.0629	N/A
Handlers-cleaners	0.7314	0.7290	0.6389	0.5556	0.0084	0.0272	0.0331	0.0564
Transport-moving	0.6893	0.7224	0.5741	0.4722	0.0791	0.0940	0.1441	0.1753
Protective-serv	0.7014	0.7404	0.4776	0.3731	0.1358	0.1481	0.2489	0.2882
Priv-house-serv	0.8231	0.8949	0.3333	0.3333	0.0204	0.0000	0.0577	0.0385
Armed-Forces	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7626	0.7476	0.6250	0.6250	0.0005	0.0016	0.0053	0.0063
Husband	0.7266	0.7162	0.3538	0.3574	0.1960	0.2127	0.3999	0.4074
Not-in-family	0.7450	0.7353	0.6161	0.6097	0.0081	0.0152	0.0451	0.0521
Unmarried	0.7044	0.7072	0.7059	0.6941	0.0041	0.0057	0.0229	0.0252
Wife	0.7402	0.7082	0.2939	0.3626	0.2264	0.2233	0.4431	0.4103
Other-relative	0.6685	0.6685	0.7857	0.7857	0.0000	0.0000	0.0085	0.0085

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7860	0.7765	0.3838	0.3917	0.0744	0.0820	0.2124	0.2161
Black	0.7868	0.7760	0.4861	0.5000	0.0211	0.0240	0.0809	0.0818
Amer-Indian-Eskimo	0.7369	0.7114	0.4545	0.4545	0.0522	0.0696	0.0952	0.1111
Asian-Pacific Islander	0.7755	0.7594	0.4167	0.4271	0.0696	0.0842	0.2033	0.2114
Other	0.7525	0.7935	0.5455	0.4545	0.0260	0.0260	0.0795	0.0909

Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7995	0.7786	0.4481	0.4879	0.0225	0.0247	0.0816	0.0791
Male	0.7734	0.7652	0.3803	0.3823	0.0962	0.1068	0.2547	0.2615

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7873	0.7776	0.3856	0.3927	0.0709	0.0789	0.2029	0.2073
Ireland	0.8125	0.6975	0.4000	0.6000	0.0000	0.0000	0.2500	0.1667
Cuba	0.8137	0.8137	0.3750	0.3750	0.0455	0.0455	0.2000	0.2000
Missing	0.8283	N/A	0.3158	N/A	0.0592	N/A	0.2297	N/A
Puerto-Rico	0.8639	0.9231	0.2500	0.2500	0.0222	0.0000	0.0816	0.0612
Philippines	0.7633	0.7203	0.4375	0.5000	0.0702	0.0877	0.1781	0.1781
China	0.8798	0.7997	0.3000	0.4000	0.0000	0.0455	0.2188	0.2188
Germany	0.8393	0.8272	0.3571	0.2857	0.0250	0.0750	0.1852	0.2407
Mexico	0.6026	0.6178	0.8333	0.8333	0.0098	0.0049	0.0142	0.0095
Peru	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0909	0.0909
India	0.7402	0.6625	0.4118	0.4118	0.1053	0.2632	0.3333	0.4167
Guatemala	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Taiwan	0.7964	0.8611	0.2857	0.2857	0.1250	0.0000	0.4000	0.3333
Vietnam	0.7262	0.8217	0.0000	0.0000	0.0909	0.0455	0.1304	0.0870

Jamaica	0.4468	0.4468	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Hungary	0.4286	0.4286	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Portugal	0.4118	0.4118	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Greece	0.5625	0.4974	0.6250	0.6250	0.1667	0.3333	0.2857	0.3571
Haiti	0.8857	0.8857	0.3333	0.3333	0.0000	0.0000	0.1000	0.1000
Nicaragua	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
South	0.8462	0.8462	0.2857	0.2857	0.0526	0.0526	0.2308	0.2308
France	0.4167	0.4167	0.3333	0.3333	0.7500	0.7500	0.7143	0.7143
Canada	0.7166	0.6485	0.5294	0.6471	0.0645	0.0645	0.2083	0.1667
El-Salvador	0.9206	0.9206	0.2500	0.2500	0.0000	0.0000	0.0857	0.0857
England	0.7286	0.7286	0.5385	0.5385	0.0400	0.0400	0.1842	0.1842
Poland	0.5567	0.5567	0.8333	0.8333	0.0000	0.0000	0.0556	0.0556
Outlying-US (Guam-USV I-etc)	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Honduras	0.4706	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Dominican- Republic	0.8857	0.7222	0.0000	0.5000	0.0556	0.0556	0.1500	0.1000
Italy	0.6813	0.7375	0.6000	0.5000	0.0556	0.0556	0.1786	0.2143
Trinidad&T obago	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000
Iran	0.5238	0.6348	0.6667	0.6667	0.2727	0.0909	0.2857	0.1429
Columbia	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0526	0.0526
Hong	0.4167	0.4167	1.0000	1.0000	0.1667	0.1667	0.1429	0.1429
Yugoslavia	0.4167	0.4167	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Ecuador	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Cambodia	0.5333	0.7083	0.7500	0.5000	0.0000	0.0000	0.1429	0.2857
Japan	0.8392	0.8873	0.2857	0.2857	0.0625	0.0000	0.2609	0.2174
Scotland	0.4444	0.4444	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Laos	0.7778	0.7778	0.5000	0.5000	0.0000	0.0000	0.1667	0.1667
Thailand	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Stage 6: Post-Mitigation ML Model (SMOTE)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood	4689	0.8347	0.7650	0.2329	0.2224	0.3773	0.1009
Elderly	890	0.8360	0.7762	0.2438	0.2393	0.3456	0.1055
Midlife	4182	0.7728	0.7576	0.3757	0.3737	0.3049	0.1804
Youth	2450	0.9853	0.6630	0.0171	0.0049	0.7857	0.0012

Detailed Group Metrics: Workclass

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Self-emp-not-inc	971	0.8033	0.7346	0.2750	0.2163	0.4644	0.0952
Private	8482	0.8518	0.7831	0.2213	0.2160	0.3468	0.0917
Self-emp-inc	411	0.7445	0.7345	0.5766	0.6180	0.1857	0.3506
Local-gov	744	0.8011	0.7531	0.2621	0.2970	0.3128	0.1585
Federal-gov	350	0.8143	0.8063	0.3886	0.4086	0.2132	0.1682
State-gov	535	0.8523	0.7988	0.2636	0.2206	0.3617	0.0711
?	713	0.9130	0.7221	0.0968	0.0743	0.5652	0.0357
Without-pay	3	1.0000	1.0000	0.0000	0.0000	0	0.0000
Never-worked	2	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8001	0.7955	0.4153	0.4346	0.2174	0.1874
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8504	0.6990	0.1576	0.1331	0.5522	0.0743
Assoc-voc	497	0.7827	0.7279	0.2656	0.2857	0.3712	0.1616
Prof-school	224	0.8661	0.8193	0.7009	0.8080	0.0191	0.4030
10th	355	0.9183	0.6560	0.0789	0.0479	0.7143	0.0275
Some-college	2683	0.8479	0.7338	0.1826	0.1625	0.4714	0.0807

7th-8th	249	0.9197	0.5942	0.0884	0.0161	0.8636	0.0044
Masters	637	0.7975	0.7939	0.5589	0.5730	0.1685	0.2456
11th	442	0.9570	0.6824	0.0475	0.0226	0.7143	0.0095
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.7836	0.7267	0.2427	0.3008	0.3261	0.1812
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9351	0.7326	0.0844	0.0455	0.6154	0.0142
Doctorate	155	0.7290	0.6668	0.6581	0.7742	0.1176	0.5660
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9702	0.7657	0.0440	0.0216	0.5922	0.0039
Married-civ-spouse	5567	0.7133	0.7113	0.4489	0.4672	0.2989	0.2767
Divorced	1682	0.9257	0.7269	0.0957	0.0511	0.6211	0.0164
Separated	373	0.9464	0.7683	0.0777	0.0456	0.5517	0.0116
Married-spouse-absent	152	0.8684	0.7237	0.1184	0.1579	0.3889	0.0970
Widowed	359	0.9331	0.6965	0.0864	0.0306	0.7097	0.0061
Married-AF-spouse	13	0.9231	0.9150	0.3846	0.3077	0.2000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9463	0.6669	0.0441	0.0401	0.6545	0.0260
Farming-fishing	370	0.8459	0.6558	0.1243	0.1324	0.5870	0.0926
Sales	1368	0.8231	0.7641	0.2617	0.2383	0.3827	0.1040
Exec-managerial	1472	0.7928	0.7928	0.5061	0.5041	0.2067	0.2077
Craft-repair	1564	0.7916	0.6944	0.2212	0.2148	0.4855	0.1297
Prof-specialty	1568	0.8106	0.8074	0.4298	0.4420	0.2062	0.1767

Machine-op-i nspct	727	0.8446	0.6738	0.1486	0.1279	0.5926	0.0792
Tech-support	394	0.8173	0.7691	0.3020	0.2411	0.4034	0.0873
Adm-clerical	1413	0.8903	0.7584	0.1331	0.1281	0.4309	0.0604
?	715	0.9133	0.7222	0.0965	0.0741	0.5652	0.0356
Handlers-clea ners	514	0.9163	0.7052	0.0700	0.0837	0.5000	0.0523
Transport-mo ving	576	0.8177	0.6836	0.1875	0.1615	0.5556	0.0962
Protective-ser v	229	0.7336	0.6875	0.2926	0.3231	0.4030	0.2099
Priv-house-se rv	52	0.9615	0.8231	0.0577	0.0577	0.3333	0.0204
Armed-Force s	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9905	0.7931	0.0126	0.0105	0.4583	0.0037
Husband	4916	0.7101	0.7084	0.4530	0.4695	0.3018	0.2800
Not-in-family	3148	0.9288	0.7512	0.0985	0.0565	0.5742	0.0162
Unmarried	1311	0.9474	0.7011	0.0648	0.0275	0.6941	0.0082
Wife	580	0.7293	0.7278	0.4517	0.4741	0.2748	0.2673
Other-relative	352	0.9602	0.7396	0.0398	0.0398	0.5000	0.0207

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8358	0.7815	0.2547	0.2468	0.3380	0.1049
Black	1186	0.9106	0.7785	0.1214	0.1062	0.4306	0.0422
Amer-Indian-Eskimo	126	0.9127	0.7719	0.0873	0.1270	0.2727	0.0696
Asian-Pac-Isl ander	369	0.8184	0.7680	0.2602	0.2737	0.3229	0.1319
Other	88	0.9091	0.7922	0.1250	0.1250	0.3636	0.0519

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9228	0.7920	0.1117	0.0951	0.4194	0.0341
Male	8154	0.8045	0.7679	0.3028	0.2999	0.3277	0.1381

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8419	0.7833	0.2430	0.2370	0.3377	0.1005
Ireland	12	0.7500	0.7333	0.4167	0.3333	0.4000	0.1429
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
?	209	0.8421	0.7976	0.2727	0.2584	0.3158	0.0987
Puerto-Rico	49	0.9796	0.9388	0.0816	0.1020	0.0000	0.0222
Philippines	73	0.8767	0.8371	0.2192	0.2877	0.1250	0.1228
China	32	0.8125	0.7681	0.3125	0.2500	0.4000	0.0909
Germany	54	0.8333	0.7778	0.2593	0.2407	0.3571	0.1000
Mexico	211	0.9573	0.5800	0.0284	0.0237	0.8333	0.0195
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.6667	0.6656	0.4722	0.4722	0.3529	0.3158
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8000	0.7964	0.4667	0.4000	0.2857	0.1250
Vietnam	23	0.8261	0.6167	0.0435	0.2174	0.0000	0.1818
Jamaica	26	0.8462	0.6232	0.1923	0.0385	0.8000	0.0000
Hungary	4	1.0000	1.0000	0.2500	0.2500	0.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.6429	0.6410	0.5714	0.5000	0.3750	0.3333
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8462	0.8045	0.2692	0.2692	0.2857	0.1053
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7708	0.7166	0.3542	0.2083	0.5294	0.0645
El-Salvador	35	1.0000	1.0000	0.1143	0.1143	0.0000	0.0000

England	38	0.7632	0.7039	0.3421	0.2105	0.5385	0.0800
Poland	18	0.7778	0.6786	0.3333	0.1111	0.6667	0.0000
Outlying-US(Guam-USVI- tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9500	0.8857	0.1000	0.1500	0.0000	0.0556
Italy	28	0.7500	0.6813	0.3571	0.1786	0.6000	0.0556
Trinidad&To bago	10	0.9000	0.8039	0.1000	0.2000	0.0000	0.1111
Iran	14	0.6429	0.5906	0.2143	0.4286	0.3333	0.3636
Columbia	19	0.9474	0.8190	0.0526	0.1053	0.0000	0.0556
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.4286	0.3000	0.2857	0.2857	1.0000	0.4000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.8261	0.7946	0.3043	0.3043	0.2857	0.1250
Scotland	5	1.0000	1.0000	0.2000	0.2000	0.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	0.7500	0.4286	0.0000	0.2500	0	0.2500

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9172	0.7913	0.1202	0.1030	0.4158	0.0373
Black + Male	600	0.8683	0.7639	0.1783	0.1567	0.4299	0.0669
White + Male	7181	0.7988	0.7664	0.3158	0.3121	0.3245	0.1443
Black + Female	586	0.9539	0.7921	0.0631	0.0546	0.4324	0.0200
Amer-Indian- Eskimo + Male	73	0.8904	0.7192	0.0685	0.1507	0.2000	0.1029
Asian-Pac-Isl ander + Female	131	0.9160	0.7729	0.1221	0.0840	0.5000	0.0261

Amer-Indian-Eskimo + Female	53	0.9434	0.8478	0.1132	0.0943	0.3333	0.0213
Asian-Pac-Islander + Male	238	0.7647	0.7438	0.3361	0.3782	0.2875	0.2089
Other + Female	26	0.9231	0.7292	0.0769	0.0769	0.5000	0.0417
Other + Male	62	0.9032	0.8050	0.1452	0.1452	0.3333	0.0566

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood + Female	1505	0.9030	0.7840	0.1389	0.1189	0.4211	0.0448
Early Adulthood + Male	3184	0.8024	0.7519	0.2773	0.2714	0.3669	0.1326
Elderly + Male	609	0.7898	0.7573	0.3169	0.3169	0.3316	0.1538
Midlife + Female	1188	0.8822	0.7861	0.1759	0.1540	0.3971	0.0582
Youth + Female	1083	0.9917	0.6979	0.0102	0.0037	0.7273	0.0009
Youth + Male	1367	0.9802	0.6488	0.0227	0.0059	0.8065	0.0015
Elderly + Female	281	0.9359	0.7781	0.0854	0.0712	0.4583	0.0272
Midlife + Male	2994	0.7295	0.7275	0.4549	0.4609	0.2907	0.2537

Comparative Group Metrics: Before vs After (SMOTE)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early Adulthood	0.7641	0.7650	0.4451	0.3773	0.0664	0.1009	0.1802	0.2224
Elderly	0.7842	0.7762	0.4147	0.3456	0.0594	0.1055	0.1876	0.2393
Midlife	0.7698	0.7576	0.3393	0.3049	0.1329	0.1804	0.3312	0.3737
Youth	0.6662	0.6630	0.7857	0.7857	0.0008	0.0012	0.0045	0.0049

Workclass

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
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Self-emp-not-inc	0.7488	0.7346	0.4757	0.4644	0.0682	0.0952	0.1936	0.2163
Private	0.7831	0.7831	0.4065	0.3468	0.0634	0.0917	0.1807	0.2160
Self-emp-inc	0.7922	0.7345	0.1983	0.1857	0.2126	0.3506	0.5523	0.6180
Local-gov	0.7478	0.7531	0.4051	0.3128	0.1129	0.1585	0.2392	0.2970
Federal-gov	0.8148	0.8063	0.2500	0.2132	0.1262	0.1682	0.3686	0.4086
State-gov	0.8142	0.7988	0.3688	0.3617	0.0482	0.0711	0.2019	0.2206
Missing	0.7330	N/A	0.5797	N/A	0.0248	N/A	0.0631	N/A
Without-pay	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Never-worked	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.8024	0.7955	0.2500	0.2174	0.1486	0.1874	0.3984	0.4346
9th	0.4918	0.4918	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
HS-grad	0.6929	0.6990	0.6282	0.5522	0.0435	0.0743	0.0952	0.1331
Assoc-voc	0.7353	0.7279	0.4697	0.3712	0.0904	0.1616	0.2072	0.2857
Prof-school	0.8358	0.8193	0.0510	0.0191	0.3134	0.4030	0.7589	0.8080
10th	0.6880	0.6560	0.7143	0.7143	0.0092	0.0275	0.0310	0.0479
Some-college	0.7357	0.7338	0.5061	0.4714	0.0625	0.0807	0.1413	0.1625
7th-8th	0.5578	0.5942	0.9091	0.8636	0.0044	0.0044	0.0120	0.0161
Masters	0.8025	0.7939	0.2079	0.1685	0.1815	0.2456	0.5228	0.5730
11th	0.6895	0.6824	0.7143	0.7143	0.0071	0.0095	0.0204	0.0226
5th-6th	0.4762	0.4762	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Assoc-acdm	0.7277	0.7267	0.4891	0.3261	0.0871	0.1812	0.1900	0.3008
Preschool	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0455	0.0455
12th	0.7640	0.7326	0.6154	0.6154	0.0000	0.0142	0.0325	0.0455
Doctorate	0.6851	0.6668	0.1373	0.1176	0.5094	0.5660	0.7419	0.7742
1st-4th	0.4945	0.4945	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7580	0.7657	0.6201	0.5922	0.0023	0.0039	0.0189	0.0216
Married-civ-spouse	0.7296	0.7113	0.3489	0.2989	0.1952	0.2767	0.3999	0.4672
Divorced	0.7244	0.7269	0.6460	0.6211	0.0105	0.0164	0.0434	0.0511
Separated	0.7585	0.7683	0.5862	0.5517	0.0087	0.0116	0.0402	0.0456
Married-spouse-absent	0.7603	0.7237	0.6111	0.3889	0.0000	0.0970	0.0461	0.1579
Widowed	0.7025	0.6965	0.7097	0.7097	0.0030	0.0061	0.0279	0.0306
Married-AF-spouse	0.3810	0.9150	1.0000	0.2000	0.0000	0.0000	0.0000	0.3077

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6221	0.6669	0.8364	0.6545	0.0034	0.0260	0.0104	0.0401
Farming-fishing	0.7095	0.6558	0.5870	0.5870	0.0432	0.0926	0.0892	0.1324
Sales	0.7677	0.7641	0.4106	0.3827	0.0832	0.1040	0.2156	0.2383
Exec-managerial	0.7960	0.7928	0.2470	0.2067	0.1596	0.2077	0.4599	0.5041
Craft-repair	0.6873	0.6944	0.5578	0.4855	0.0952	0.1297	0.1720	0.2148
Prof-specialty	0.8081	0.8074	0.2537	0.2062	0.1353	0.1767	0.3980	0.4420
Machine-op-inspct	0.6410	0.6738	0.7407	0.5926	0.0323	0.0792	0.0660	0.1279
Tech-support	0.7700	0.7691	0.4286	0.4034	0.0691	0.0873	0.2208	0.2411
Adm-clerical	0.7705	0.7584	0.4521	0.4309	0.0441	0.0604	0.1111	0.1281
Missing	0.7331	N/A	0.5797	N/A	0.0248	N/A	0.0629	N/A
Handlers-cleaners	0.7314	0.7052	0.6389	0.5000	0.0084	0.0523	0.0331	0.0837
Transport-moving	0.6893	0.6836	0.5741	0.5556	0.0791	0.0962	0.1441	0.1615
Protective-serv	0.7014	0.6875	0.4776	0.4030	0.1358	0.2099	0.2489	0.3231
Priv-house-serv	0.8231	0.8231	0.3333	0.3333	0.0204	0.0204	0.0577	0.0577

Armed-Forces	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
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Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7626	0.7931	0.6250	0.4583	0.0005	0.0037	0.0053	0.0105
Husband	0.7266	0.7084	0.3538	0.3018	0.1960	0.2800	0.3999	0.4695
Not-in-family	0.7450	0.7512	0.6161	0.5742	0.0081	0.0162	0.0451	0.0565
Unmarried	0.7044	0.7011	0.7059	0.6941	0.0041	0.0082	0.0229	0.0275
Wife	0.7402	0.7278	0.2939	0.2748	0.2264	0.2673	0.4431	0.4741
Other-relative	0.6685	0.7396	0.7857	0.5000	0.0000	0.0207	0.0085	0.0398

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7860	0.7815	0.3838	0.3380	0.0744	0.1049	0.2124	0.2468
Black	0.7868	0.7785	0.4861	0.4306	0.0211	0.0422	0.0809	0.1062
Amer-Indian-Eskimo	0.7369	0.7719	0.4545	0.2727	0.0522	0.0696	0.0952	0.1270
Asian-Pac-Islander	0.7755	0.7680	0.4167	0.3229	0.0696	0.1319	0.2033	0.2737
Other	0.7525	0.7922	0.5455	0.3636	0.0260	0.0519	0.0795	0.1250

Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7995	0.7920	0.4481	0.4194	0.0225	0.0341	0.0816	0.0951
Male	0.7734	0.7679	0.3803	0.3277	0.0962	0.1381	0.2547	0.2999

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7873	0.7833	0.3856	0.3377	0.0709	0.1005	0.2029	0.2370
Ireland	0.8125	0.7333	0.4000	0.4000	0.0000	0.1429	0.2500	0.3333

Cuba	0.8137	0.8137	0.3750	0.3750	0.0455	0.0455	0.2000	0.2000
Missing	0.8283	N/A	0.3158	N/A	0.0592	N/A	0.2297	N/A
Puerto-Rico	0.8639	0.9388	0.2500	0.0000	0.0222	0.0222	0.0816	0.1020
Philippines	0.7633	0.8371	0.4375	0.1250	0.0702	0.1228	0.1781	0.2877
China	0.8798	0.7681	0.3000	0.4000	0.0000	0.0909	0.2188	0.2500
Germany	0.8393	0.7778	0.3571	0.3571	0.0250	0.1000	0.1852	0.2407
Mexico	0.6026	0.5800	0.8333	0.8333	0.0098	0.0195	0.0142	0.0237
Peru	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0909	0.0909
India	0.7402	0.6656	0.4118	0.3529	0.1053	0.3158	0.3333	0.4722
Guatemala	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Taiwan	0.7964	0.7964	0.2857	0.2857	0.1250	0.1250	0.4000	0.4000
Vietnam	0.7262	0.6167	0.0000	0.0000	0.0909	0.1818	0.1304	0.2174
Jamaica	0.4468	0.6232	1.0000	0.8000	0.0000	0.0000	0.0000	0.0385
Hungary	0.4286	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.2500
Portugal	0.4118	0.4118	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Greece	0.5625	0.6410	0.6250	0.3750	0.1667	0.3333	0.2857	0.5000
Haiti	0.8857	0.8857	0.3333	0.3333	0.0000	0.0000	0.1000	0.1000
Nicaragua	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
South	0.8462	0.8045	0.2857	0.2857	0.0526	0.1053	0.2308	0.2692
France	0.4167	0.4167	0.3333	0.3333	0.7500	0.7500	0.7143	0.7143
Canada	0.7166	0.7166	0.5294	0.5294	0.0645	0.0645	0.2083	0.2083
El-Salvador	0.9206	1.0000	0.2500	0.0000	0.0000	0.0000	0.0857	0.1143
England	0.7286	0.7039	0.5385	0.5385	0.0400	0.0800	0.1842	0.2105
Poland	0.5567	0.6786	0.8333	0.6667	0.0000	0.0000	0.0556	0.1111
Outlying-US (Guam-USV I-etc)	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Honduras	0.4706	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Dominican- Republic	0.8857	0.8857	0.0000	0.0000	0.0556	0.0556	0.1500	0.1500
Italy	0.6813	0.6813	0.6000	0.6000	0.0556	0.0556	0.1786	0.1786
Trinidad&T obago	1.0000	0.8039	0.0000	0.0000	0.0000	0.1111	0.1000	0.2000
Iran	0.5238	0.5906	0.6667	0.3333	0.2727	0.3636	0.2857	0.4286

Columbia	1.0000	0.8190	0.0000	0.0000	0.0000	0.0556	0.0526	0.1053
Hong	0.4167	0.4167	1.0000	1.0000	0.1667	0.1667	0.1429	0.1429
Yugoslavia	0.4167	0.3000	1.0000	1.0000	0.0000	0.4000	0.0000	0.2857
Ecuador	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Cambodia	0.5333	0.7083	0.7500	0.5000	0.0000	0.0000	0.1429	0.2857
Japan	0.8392	0.7946	0.2857	0.2857	0.0625	0.1250	0.2609	0.3043
Scotland	0.4444	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.2000
Laos	0.7778	0.7778	0.5000	0.5000	0.0000	0.0000	0.1667	0.1667
Thailand	1.0000	0.4286	0	0	0.0000	0.2500	0.0000	0.2500

Stage 6: Post-Mitigation ML Model (AIF360 Reweighing)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood	4689	0.8405	0.7554	0.2329	0.1774	0.4615	0.0678
Elderly	890	0.8539	0.7850	0.2438	0.1899	0.4101	0.0609
Midlife	4182	0.7848	0.7648	0.3757	0.3331	0.3431	0.1383
Youth	2450	0.9853	0.6630	0.0171	0.0049	0.7857	0.0012

Detailed Group Metrics: Workclass

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Self-emp-not-inc	971	0.8136	0.7378	0.2750	0.1874	0.4981	0.0682
Private	8482	0.8590	0.7807	0.2213	0.1812	0.4092	0.0648
Self-emp-inc	411	0.7908	0.7869	0.5766	0.5572	0.1983	0.2241
Local-gov	744	0.8051	0.7415	0.2621	0.2419	0.4103	0.1184
Federal-gov	350	0.8114	0.7981	0.3886	0.3543	0.2868	0.1262
State-gov	535	0.8486	0.7880	0.2636	0.2019	0.4043	0.0609
?	713	0.9173	0.7210	0.0968	0.0645	0.5942	0.0280
Without-pay	3	1.0000	1.0000	0.0000	0.0000	0	0.0000

Never-worked	2	1.0000	1.0000	0.0000	0.0000	0	0.0000
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Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.7948	0.7869	0.4153	0.3926	0.2744	0.1561
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8671	0.7009	0.1576	0.0970	0.6139	0.0429
Assoc-voc	497	0.7948	0.7132	0.2656	0.2012	0.5076	0.0959
Prof-school	224	0.8705	0.8284	0.7009	0.7946	0.0255	0.3731
10th	355	0.9380	0.6942	0.0789	0.0282	0.7143	0.0061
Some-college	2683	0.8528	0.7288	0.1826	0.1413	0.5163	0.0648
7th-8th	249	0.9157	0.5578	0.0884	0.0120	0.9091	0.0044
Masters	637	0.7834	0.7821	0.5589	0.5181	0.2303	0.1993
11th	442	0.9593	0.7082	0.0475	0.0249	0.6667	0.0095
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8259	0.7409	0.2427	0.1847	0.4783	0.0767
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7355	0.6811	0.6581	0.7548	0.1275	0.5283
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9692	0.7507	0.0440	0.0197	0.6257	0.0033
Married-civ-spouse	5567	0.7291	0.7226	0.4489	0.3982	0.3581	0.1998
Divorced	1682	0.9269	0.7186	0.0957	0.0440	0.6522	0.0118
Separated	373	0.9437	0.7517	0.0777	0.0429	0.5862	0.0116
Married-spouse-absent	152	0.9342	0.7897	0.1184	0.0526	0.5556	0.0000
Widowed	359	0.9359	0.7025	0.0864	0.0279	0.7097	0.0030

Married-AF-spouse	13	0.6923	0.5667	0.3846	0.0769	0.8000	0.0000
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Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9567	0.6240	0.0441	0.0152	0.8182	0.0075
Farming-fishing	370	0.8784	0.6947	0.1243	0.1000	0.5870	0.0556
Sales	1368	0.8311	0.7637	0.2617	0.2039	0.4330	0.0752
Exec-managerial	1472	0.7880	0.7879	0.5061	0.4640	0.2510	0.1719
Craft-repair	1564	0.8037	0.6854	0.2212	0.1656	0.5694	0.0903
Prof-specialty	1568	0.8112	0.8055	0.4298	0.3980	0.2567	0.1376
Machine-oper-inspct	727	0.8597	0.6457	0.1486	0.0743	0.7222	0.0388
Tech-support	394	0.8299	0.7805	0.3020	0.2234	0.4118	0.0655
Adm-clerical	1413	0.8946	0.7515	0.1331	0.1083	0.4894	0.0465
?	715	0.9175	0.7210	0.0965	0.0643	0.5942	0.0279
Handlers-cleaners	514	0.9416	0.7167	0.0700	0.0389	0.6389	0.0146
Transport-moving	576	0.8212	0.6740	0.1875	0.1406	0.6019	0.0812
Protective-serv	229	0.7511	0.6895	0.2926	0.2620	0.4776	0.1543
Priv-house-serv	52	0.9615	0.8231	0.0577	0.0577	0.3333	0.0204
Armed-Forces	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9916	0.7626	0.0126	0.0053	0.6250	0.0005
Husband	4916	0.7276	0.7215	0.4530	0.3987	0.3606	0.1993
Not-in-family	3148	0.9298	0.7393	0.0985	0.0467	0.6194	0.0102
Unmarried	1311	0.9497	0.7023	0.0648	0.0236	0.7059	0.0049
Wife	580	0.7224	0.7191	0.4517	0.4397	0.3206	0.2421

Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000
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Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8436	0.7816	0.2547	0.2123	0.3902	0.0765
Black	1186	0.9140	0.7636	0.1214	0.0809	0.5208	0.0259
Amer-Indian-Eskimo	126	0.9127	0.7558	0.0873	0.1111	0.3636	0.0609
Asian-Pac-Islander	369	0.8347	0.7659	0.2602	0.1978	0.4375	0.0696
Other	88	0.9205	0.7721	0.1250	0.0682	0.5455	0.0130

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9263	0.7892	0.1117	0.0818	0.4636	0.0247
Male	8154	0.8142	0.7689	0.3028	0.2544	0.3868	0.0985

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8497	0.7833	0.2430	0.2033	0.3909	0.0731
Ireland	12	0.7500	0.6975	0.4167	0.1667	0.6000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
?	209	0.8469	0.7978	0.2727	0.2344	0.3509	0.0789
Puerto-Rico	49	0.9592	0.8639	0.0816	0.0816	0.2500	0.0222
Philippines	73	0.8219	0.7203	0.2192	0.1781	0.5000	0.0877
China	32	0.8750	0.8333	0.3125	0.1875	0.4000	0.0000
Germany	54	0.8519	0.7974	0.2593	0.2222	0.3571	0.0750
Mexico	211	0.9716	0.6178	0.0284	0.0095	0.8333	0.0049
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7222	0.7078	0.4722	0.3056	0.4706	0.1053
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000

Taiwan	15	0.8000	0.7964	0.4667	0.4000	0.2857	0.1250
Vietnam	23	0.9565	0.8217	0.0435	0.0870	0.0000	0.0455
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.7333	0.2500	0.5000	0.0000	0.3333
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.5000	0.4759	0.5714	0.2143	0.7500	0.1667
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.9231	0.8917	0.2692	0.1923	0.2857	0.0000
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7500	0.6667	0.3542	0.1458	0.6471	0.0323
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7895	0.7286	0.3421	0.1842	0.5385	0.0400
Poland	18	0.6667	0.5179	0.3333	0.1111	0.8333	0.0833
Outlying-US(Guam-USVI-e tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9000	0.7222	0.1000	0.1000	0.5000	0.0556
Italy	28	0.7500	0.6813	0.3571	0.1786	0.6000	0.0556
Trinidad&To bago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	1.0000	1.0000	0.0526	0.0526	0.0000	0.0000
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.5714	0.5333	0.5714	0.1429	0.7500	0.0000
Japan	23	0.9130	0.8873	0.3043	0.2174	0.2857	0.0000
Scotland	5	0.8000	0.4444	0.2000	0.0000	1.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9218	0.7915	0.1202	0.0892	0.4541	0.0268
Black + Male	600	0.8750	0.7491	0.1783	0.1133	0.5327	0.0365
White + Male	7181	0.8081	0.7680	0.3158	0.2682	0.3792	0.1054
Black + Female	586	0.9539	0.7801	0.0631	0.0478	0.4865	0.0164
Amer-Indian-Eskimo + Male	73	0.8904	0.7192	0.0685	0.1507	0.2000	0.1029
Asian-Pac-Islander + Female	131	0.9084	0.7248	0.1221	0.0611	0.6250	0.0174
Amer-Indian-Eskimo + Female	53	0.9434	0.8179	0.1132	0.0566	0.5000	0.0000
Asian-Pac-Islander + Male	238	0.7941	0.7570	0.3361	0.2731	0.4000	0.1076
Other + Female	26	0.9231	0.7292	0.0769	0.0769	0.5000	0.0417
Other + Male	62	0.9194	0.7852	0.1452	0.0645	0.5556	0.0000

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early Adulthood + Female	1505	0.9043	0.7771	0.1389	0.1056	0.4641	0.0363
Early Adulthood + Male	3184	0.8103	0.7432	0.2773	0.2114	0.4609	0.0856
Elderly + Male	609	0.8112	0.7675	0.3169	0.2496	0.4041	0.0889
Midlife + Female	1188	0.8889	0.7839	0.1759	0.1271	0.4545	0.0378
Youth + Female	1083	0.9926	0.7481	0.0102	0.0046	0.6364	0.0009
Youth + Male	1367	0.9795	0.6264	0.0227	0.0051	0.8387	0.0015
Elderly + Female	281	0.9466	0.8027	0.0854	0.0605	0.4583	0.0156
Midlife + Male	2994	0.7435	0.7391	0.4549	0.4148	0.3260	0.1985

Comparative Group Metrics: Before vs After (AIF360 Reweighing)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early Adulthood	0.7641	0.7554	0.4451	0.4615	0.0664	0.0678	0.1802	0.1774
Elderly	0.7842	0.7850	0.4147	0.4101	0.0594	0.0609	0.1876	0.1899
Midlife	0.7698	0.7648	0.3393	0.3431	0.1329	0.1383	0.3312	0.3331
Youth	0.6662	0.6630	0.7857	0.7857	0.0008	0.0012	0.0045	0.0049

Workclass

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Self-emp-not-inc	0.7488	0.7378	0.4757	0.4981	0.0682	0.0682	0.1936	0.1874
Private	0.7831	0.7807	0.4065	0.4092	0.0634	0.0648	0.1807	0.1812
Self-emp-inc	0.7922	0.7869	0.1983	0.1983	0.2126	0.2241	0.5523	0.5572
Local-gov	0.7478	0.7415	0.4051	0.4103	0.1129	0.1184	0.2392	0.2419
Federal-gov	0.8148	0.7981	0.2500	0.2868	0.1262	0.1262	0.3686	0.3543
State-gov	0.8142	0.7880	0.3688	0.4043	0.0482	0.0609	0.2019	0.2019
Missing	0.7330	N/A	0.5797	N/A	0.0248	N/A	0.0631	N/A
Without-pay	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Never-worked	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.8024	0.7869	0.2500	0.2744	0.1486	0.1561	0.3984	0.3926
9th	0.4918	0.4918	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
HS-grad	0.6929	0.7009	0.6282	0.6139	0.0435	0.0429	0.0952	0.0970
Assoc-voc	0.7353	0.7132	0.4697	0.5076	0.0904	0.0959	0.2072	0.2012
Prof-school	0.8358	0.8284	0.0510	0.0255	0.3134	0.3731	0.7589	0.7946
10th	0.6880	0.6942	0.7143	0.7143	0.0092	0.0061	0.0310	0.0282
Some-college	0.7357	0.7288	0.5061	0.5163	0.0625	0.0648	0.1413	0.1413

7th-8th	0.5578	0.5578	0.9091	0.9091	0.0044	0.0044	0.0120	0.0120
Masters	0.8025	0.7821	0.2079	0.2303	0.1815	0.1993	0.5228	0.5181
11th	0.6895	0.7082	0.7143	0.6667	0.0071	0.0095	0.0204	0.0249
5th-6th	0.4762	0.4762	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Assoc-acd m	0.7277	0.7409	0.4891	0.4783	0.0871	0.0767	0.1900	0.1847
Preschool	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0455	0.0455
12th	0.7640	0.7640	0.6154	0.6154	0.0000	0.0000	0.0325	0.0325
Doctorate	0.6851	0.6811	0.1373	0.1275	0.5094	0.5283	0.7419	0.7548
1st-4th	0.4945	0.4945	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7580	0.7507	0.6201	0.6257	0.0023	0.0033	0.0189	0.0197
Married-civ-spouse	0.7296	0.7226	0.3489	0.3581	0.1952	0.1998	0.3999	0.3982
Divorced	0.7244	0.7186	0.6460	0.6522	0.0105	0.0118	0.0434	0.0440
Separated	0.7585	0.7517	0.5862	0.5862	0.0087	0.0116	0.0402	0.0429
Married-spouse-absent	0.7603	0.7897	0.6111	0.5556	0.0000	0.0000	0.0461	0.0526
Widowed	0.7025	0.7025	0.7097	0.7097	0.0030	0.0030	0.0279	0.0279
Married-AF-spouse	0.3810	0.5667	1.0000	0.8000	0.0000	0.0000	0.0000	0.0769

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6221	0.6240	0.8364	0.8182	0.0034	0.0075	0.0104	0.0152
Farming-fishing	0.7095	0.6947	0.5870	0.5870	0.0432	0.0556	0.0892	0.1000
Sales	0.7677	0.7637	0.4106	0.4330	0.0832	0.0752	0.2156	0.2039
Exec-managerial	0.7960	0.7879	0.2470	0.2510	0.1596	0.1719	0.4599	0.4640
Craft-repair	0.6873	0.6854	0.5578	0.5694	0.0952	0.0903	0.1720	0.1656

Prof-specialty	0.8081	0.8055	0.2537	0.2567	0.1353	0.1376	0.3980	0.3980
Machine-op-inspct	0.6410	0.6457	0.7407	0.7222	0.0323	0.0388	0.0660	0.0743
Tech-support	0.7700	0.7805	0.4286	0.4118	0.0691	0.0655	0.2208	0.2234
Adm-clerical	0.7705	0.7515	0.4521	0.4894	0.0441	0.0465	0.1111	0.1083
Missing	0.7331	N/A	0.5797	N/A	0.0248	N/A	0.0629	N/A
Handlers-cleaners	0.7314	0.7167	0.6389	0.6389	0.0084	0.0146	0.0331	0.0389
Transport-moving	0.6893	0.6740	0.5741	0.6019	0.0791	0.0812	0.1441	0.1406
Protective-serv	0.7014	0.6895	0.4776	0.4776	0.1358	0.1543	0.2489	0.2620
Priv-house-serv	0.8231	0.8231	0.3333	0.3333	0.0204	0.0204	0.0577	0.0577
Armed-Forces	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7626	0.7626	0.6250	0.6250	0.0005	0.0005	0.0053	0.0053
Husband	0.7266	0.7215	0.3538	0.3606	0.1960	0.1993	0.3999	0.3987
Not-in-family	0.7450	0.7393	0.6161	0.6194	0.0081	0.0102	0.0451	0.0467
Unmarried	0.7044	0.7023	0.7059	0.7059	0.0041	0.0049	0.0229	0.0236
Wife	0.7402	0.7191	0.2939	0.3206	0.2264	0.2421	0.4431	0.4397
Other-relative	0.6685	0.6685	0.7857	0.7857	0.0000	0.0000	0.0085	0.0085

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7860	0.7816	0.3838	0.3902	0.0744	0.0765	0.2124	0.2123
Black	0.7868	0.7636	0.4861	0.5208	0.0211	0.0259	0.0809	0.0809
Amer-Indian-Eskimo	0.7369	0.7558	0.4545	0.3636	0.0522	0.0609	0.0952	0.1111
Asian-Pac-Islander	0.7755	0.7659	0.4167	0.4375	0.0696	0.0696	0.2033	0.1978

Other	0.7525	0.7721	0.5455	0.5455	0.0260	0.0130	0.0795	0.0682
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Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7995	0.7892	0.4481	0.4636	0.0225	0.0247	0.0816	0.0818
Male	0.7734	0.7689	0.3803	0.3868	0.0962	0.0985	0.2547	0.2544

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7873	0.7833	0.3856	0.3909	0.0709	0.0731	0.2029	0.2033
Ireland	0.8125	0.6975	0.4000	0.6000	0.0000	0.0000	0.2500	0.1667
Cuba	0.8137	0.8137	0.3750	0.3750	0.0455	0.0455	0.2000	0.2000
Missing	0.8283	N/A	0.3158	N/A	0.0592	N/A	0.2297	N/A
Puerto-Rico	0.8639	0.8639	0.2500	0.2500	0.0222	0.0222	0.0816	0.0816
Philippines	0.7633	0.7203	0.4375	0.5000	0.0702	0.0877	0.1781	0.1781
China	0.8798	0.8333	0.3000	0.4000	0.0000	0.0000	0.2188	0.1875
Germany	0.8393	0.7974	0.3571	0.3571	0.0250	0.0750	0.1852	0.2222
Mexico	0.6026	0.6178	0.8333	0.8333	0.0098	0.0049	0.0142	0.0095
Peru	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0909	0.0909
India	0.7402	0.7078	0.4118	0.4706	0.1053	0.1053	0.3333	0.3056
Guatemala	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Taiwan	0.7964	0.7964	0.2857	0.2857	0.1250	0.1250	0.4000	0.4000
Vietnam	0.7262	0.8217	0.0000	0.0000	0.0909	0.0455	0.1304	0.0870
Jamaica	0.4468	0.4468	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Hungary	0.4286	0.7333	1.0000	0.0000	0.0000	0.3333	0.0000	0.5000
Portugal	0.4118	0.4118	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Greece	0.5625	0.4759	0.6250	0.7500	0.1667	0.1667	0.2857	0.2143
Haiti	0.8857	0.8857	0.3333	0.3333	0.0000	0.0000	0.1000	0.1000
Nicaragua	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
South	0.8462	0.8917	0.2857	0.2857	0.0526	0.0000	0.2308	0.1923
France	0.4167	0.4167	0.3333	0.3333	0.7500	0.7500	0.7143	0.7143

Canada	0.7166	0.6667	0.5294	0.6471	0.0645	0.0323	0.2083	0.1458
El-Salvador	0.9206	0.9206	0.2500	0.2500	0.0000	0.0000	0.0857	0.0857
England	0.7286	0.7286	0.5385	0.5385	0.0400	0.0400	0.1842	0.1842
Poland	0.5567	0.5179	0.8333	0.8333	0.0000	0.0833	0.0556	0.1111
Outlying-US (Guam-USV I-etc)	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Honduras	0.4706	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Dominican- Republic	0.8857	0.7222	0.0000	0.5000	0.0556	0.0556	0.1500	0.1000
Italy	0.6813	0.6813	0.6000	0.6000	0.0556	0.0556	0.1786	0.1786
Trinidad&T obago	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000
Iran	0.5238	0.5238	0.6667	0.6667	0.2727	0.2727	0.2857	0.2857
Columbia	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0526	0.0526
Hong	0.4167	0.4167	1.0000	1.0000	0.1667	0.1667	0.1429	0.1429
Yugoslavia	0.4167	0.4167	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Ecuador	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Cambodia	0.5333	0.5333	0.7500	0.7500	0.0000	0.0000	0.1429	0.1429
Japan	0.8392	0.8873	0.2857	0.2857	0.0625	0.0000	0.2609	0.2174
Scotland	0.4444	0.4444	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Laos	0.7778	0.7778	0.5000	0.5000	0.0000	0.0000	0.1667	0.1667
Thailand	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000